Statistics for Intelligent Systems Project II

Goal: The goal of the project is to show proficiency in using basic concepts from inductive statistics.

Description: As we do not have the resources to conduct an full statistical study we can still simulate that, by choosing a data set, dividing it into two parts, one we have full access to from the start and the second that will be used for tests. Because of this, when choosing a data set, please look for rather larger datasets (any of size smaller than 100 should be discarded). For the project we can choose one of three types methods to work with: We can either test a selected hypothesis; try prediction based on a linear regression model, or Anova depending on the type of data chosen; or try prediction based on a time series. Depending on the selected statistical method we are expected to prepare the data and conduct the reasoning in a different manner:

- **Hypothesis testing:** In this case we should divide the data given into roughly even parts. We describe the part that we have full knowledge about and all of its interesting parameters, and based on that and the additional knowledge we have about the data set, we formulate a null hypothesis and an alternative hypothesis. We use the second part of data to decide between the null and the alternative hypothesis.

- **Linear regression or Anova:** In this part we divide the data in two parts of size roughly about 90% of the original dataset (this part is used to create the model) and the second part from the rest (which will be used to check the validity of the model). Based on the first describe what are going to be the dependent and independent variables and create the model you will use for predictions. Now compare the data you have in the second set with the predictions your model creates. Treat the differences between the predictions and the actual values as a random variable and describe their mean, variance, histogram and other descriptive statistics.

- **Time series analysis:** This case is similar to the previous one, the only difference is in the models. Please divide the data so the first 90% of it is in the first set. Analyse that time series and use a model to predict the last 10% of data. Compare your predictions with the actual data and describe the basic statistics in those differences.

In all cases please end with a comment whether your ideas had good basis, and what are future paths you would propose to analyze to build up on the results.

Apart from the report you need to prepare a short presentation (about 3 to 4 minutes) in which you will summarize all of the information given in your report, which you will give during class on the 15.06.2026 and 22.06.2026.

Possible sources:

- `http://www.kaggle.com` one of the largest sites with access to public datasets
- `https://catalog.data.gov/dataset` datasets made available by the us government
- `https://bdl.stat.gov.pl/bdl/start` a database containing information gathered by Polish Main Statistics Institution (GUS), although retriving information is a bit more challenging here